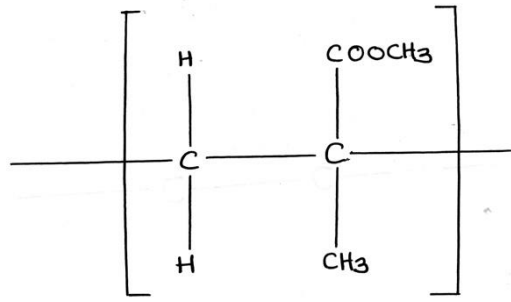


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
12.	(a)	(i)		electrophilic addition		1				1
		(ii)		acidified (potassium) dichromate / H^+ , $\text{Cr}_2\text{O}_7^{2-}$ / acidified potassium manganate(VII) / H^+ , MnO_4^-	1				1	1
		(iii)		ZnBr_2			1			1
	(b)	(i)		$-\text{CN}$		1				1
		(ii)		HCN is added across the $\text{C}=\text{O}$ bond		1				1
		(iii)		dilute sulfuric acid / $\text{H}_2\text{SO}_4(\text{aq})$ / dilute hydrochloric acid/ $\text{HCl}(\text{aq})$ neutral answer – water	1					1
		(iv)		dehydrating agent / removes water	1				1	1
		(v)				1				1